

Fiber-Optic Cables — Guided Notes ANSWER KEY

Unit 1.3.2 | CompTIA Network+ N10-009 Obj. 1.3 | N10-008 1.3

For instructor use / distribute after students complete the notes.

Question 1

Fiber-optic cable transmits data as pulses of _____ instead of electrical signals. This makes it immune to _____ and safe to run between buildings without risk of _____.

Answer: Light; electromagnetic interference (EMI); ground loops or lightning damage.

Question 2

Multimode fiber has a _____ (larger/smaller) core than single-mode. Its light source is a _____, which is cheaper than the _____ used in single-mode. The main distance limitation of multimode is caused by _____.

Answer: Larger; LED; laser; modal dispersion.

Question 3

Single-mode fiber has a core of approximately _____ microns and supports runs of up to _____ km without a repeater. Its jacket color is _____. Multimode jacket colors are _____ or _____.

Answer: 8-9 microns; 80 km; yellow; orange or aqua.

Question 4

Match each fiber connector to its description:

SC → _____

LC → _____

ST → _____

Answer: SC → square, push-pull, found on older installations; LC → small form-factor, latch clip, dominates modern SFP/SFP+ installations; ST → round, bayonet twist-lock, legacy multimode.

Question 5

Fiber connections always come in pairs because one strand _____ and the other _____. Swapping these two strands is a common mistake that mixes up _____ and _____.

Answer: Transmits; receives; TX; RX.

Question 6

UPC connectors have a _____ housing. APC connectors have a _____ housing and polish the endface at a _____ angle. These two types are physically _____ and must never be mated.

Answer: Blue; green; 8-degree; incompatible.

Question 7

Exceeding a fiber cable's minimum _____ causes macrobending loss. Contaminated fiber endfaces must be cleaned with _____ before every connection. Unused connectors and ports should always

be _____.

Answer: Bend radius; lint-free wipes and isopropyl alcohol (or one-click cleaners); capped.

Question 8 — Real World Application

REAL WORLD: A backbone fiber link between two wiring closets starts throwing CRC errors and negotiates at a lower speed. Both ends use LC connectors with blue boots. Walk through your troubleshooting steps in order.

Answer: 1. Inspect run for bend radius violations. 2. Clean both endfaces with a one-click cleaner or lint-free wipe + IPA. 3. Use a fiber light meter to check power levels at the receive end. 4. Inspect connectors under a fiber microscope for contamination or damage. 5. Confirm both are UPC (blue = correct). 6. Test with a known-good patch cable to isolate permanent link vs patch.